

REDESDALE CAMP, United Kingdom

WMO: 032300

Lat: 55.2850N

Lon: 2.2794W

Elev: 212

StdP: 98.80

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.8	-4.2	-7.4	2.1	-4.7	-6.0	2.3	-3.1	14.9	7.5	13.6	6.7	1.3	0	0.649

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	22.8	17.2	21.0	16.1	19.3	15.1	17.9	21.7	16.9	20.0	15.9	18.4	3.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.4	12.0	19.4	15.6	11.3	18.3	14.7	10.7	17.3	51.2	21.8	48.1	20.1	45.3	18.5	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.6	10.1	8.7	-9.3	25.8	2.8	1.8	-11.4	27.1	-13.0	28.2	-14.6	29.2	-16.7	30.5		
(5)			-9.4	19.9	2.8	1.2	-11.4	20.8	-13.1	21.5	-14.6	22.2	-16.7	23.0	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	2.8	3.1	4.3	6.5	9.2	12.0	13.8	13.5	11.4	8.3	5.0	3.1
	DBStd	4.90	2.93	2.92	2.90	2.79	2.92	2.41	2.38	2.30	2.37	2.82	3.12	3.62
	HDD10.0	1203	222	193	176	108	49	8	1	1	11	68	151	216
	HDD18.3	3851	481	426	434	353	283	192	141	149	208	312	399	473
	CDD10.0	397	0	0	1	4	26	67	119	111	54	14	2	0
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0
	CDH23.3	38	0	0	0	0	1	4	20	12	1	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	1	0	0	0	0
Wind	WSAvg	3.6	4.1	4.3	4.1	3.6	3.6	3.2	3.0	2.9	3.1	3.4	3.6	3.8
Precipitation	PrecAvg	1054	114	89	71	72	68	72	78	85	74	105	112	116
	PrecMax	1361	229	188	127	155	112	173	148	188	141	183	232	298
	PrecMin	742	19	22	24	17	17	19	26	14	13	37	36	37
	PrecStd	143	46	48	26	35	28	40	34	40	32	42	44	52
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.9	11.3	14.4	18.7	22.3	23.7	25.6	25.2	21.5	16.7	13.6	11.8
		MCWB	9.9	9.2	9.4	12.5	15.9	17.3	18.4	17.8	16.8	14.0	12.4	10.8
	2%	DB	9.5	9.7	11.6	15.5	19.5	21.2	23.2	21.5	19.0	14.9	12.0	10.5
		MCWB	8.4	8.4	8.7	11.0	14.0	15.9	17.5	17.0	15.1	13.1	10.9	9.6
	5%	DB	8.4	8.8	10.2	13.3	17.1	19.1	21.1	19.5	17.3	13.9	10.9	9.5
		MCWB	7.5	7.5	8.1	9.6	12.7	14.5	16.4	15.8	14.2	12.5	9.8	8.6
	10%	DB	7.4	7.8	9.1	11.8	14.9	17.3	19.0	18.0	15.9	12.8	9.9	8.4
		MCWB	6.5	6.6	7.3	8.9	11.3	13.5	15.4	14.9	13.4	11.3	8.8	7.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	9.8	10.7	13.3	16.9	18.5	19.5	18.8	17.7	14.7	12.7	11.0
		MCDB	10.7	10.9	13.3	17.3	21.6	22.5	24.1	22.3	20.8	15.7	13.5	11.6
	2%	WB	8.7	8.7	9.4	11.5	14.5	16.5	18.1	17.5	16.0	13.6	11.2	9.8
		MCDB	9.3	9.6	10.9	14.6	18.4	20.1	22.0	20.7	17.9	14.5	11.8	10.4
	5%	WB	7.7	7.8	8.4	10.4	13.1	15.1	16.9	16.5	14.8	12.7	10.0	8.8
		MCDB	8.4	8.7	9.8	12.8	16.3	18.2	20.3	18.9	16.5	13.8	10.7	9.5
	10%	WB	6.6	6.7	7.5	9.3	11.8	14.0	15.8	15.6	13.8	11.6	8.9	7.6
		MCDB	7.4	7.7	9.0	11.5	14.6	16.6	18.5	17.5	15.5	12.6	9.7	8.4
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.9	7.0	8.8	9.3	8.9	9.0	8.7	8.4	7.1	5.9	5.4
		MCDBR	5.8	6.7	8.8	11.9	13.2	12.8	12.9	11.1	10.8	8.0	6.4	5.9
		MCWBR	5.5	5.6	6.2	7.8	8.4	8.2	7.7	6.9	7.3	6.4	5.9	5.8
	5% WB	MCDBR	5.6	6.2	7.9	10.7	12.0	11.6	11.6	9.7	9.1	7.2	6.4	5.9
		MCWBR	5.5	5.6	6.2	7.4	7.9	7.9	7.4	6.4	6.8	6.2	6.0	5.9
Clear-Sky Solar Irradiance	taub	0.310	0.325	0.355	0.377	0.376	0.376	0.376	0.366	0.349	0.333	0.308	0.293	
	taud	2.284	2.372	2.311	2.279	2.323	2.347	2.350	2.390	2.447	2.447	2.354	2.254	
	Ebn at Noon	608	752	809	843	866	869	860	847	814	735	612	533	
	Edh at Noon	56	73	97	115	117	116	113	102	85	67	52	46	
All-Sky Solar Radiation	RadAvg	0.55	1.19	2.27	3.68	4.79	4.71	4.53	3.69	2.63	1.44	0.70	0.41	
	RadStd	0.08	0.12	0.24	0.38	0.34	0.44	0.46	0.28	0.22	0.16	0.06	0.04	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page